THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the three exterior angles of a triangle (one at each vertex) is: 1

(a) 360° **(b)** 540° **(c)** 180° **(d)** 270°

2. The angles of a triangle are $(3x)^\circ$, $(2x+10)^\circ$ and $(4x-10)^\circ$. The largest angle is: 1

(a) 80° **(b)** 60° **(c)** 70° **(d)** 90°

3. Sides 13 cm, 6 cm and 8 cm are given. Which statement is true? 1

(a) no triangle can be formed **(b)** only an equilateral triangle is possible

(c) a triangle can be formed **(d)** only a right triangle is possible

4. In an isosceles triangle with perimeter 50 cm the equal sides are 18 cm each. The base (cm) is: 1

(a) 16 **(b)** 14 **(c)** 13 **(d)** 15

5. The two acute angles of a right triangle are in the ratio $2:3$. The smaller acute angle is: 1

(a) 26° **(b)** 36° **(c)** 46° **(d)** 56°

6. The number of isosceles triangles with whole-number sides and perimeter 15 cm (equilateral included) is: 1

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

7. The number of triangles with whole-number sides (in cm) and perimeter 14 cm is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

8. The exterior angle at the base of an isosceles triangle is 140°. The apex angle is: 1

(a) 90° **(b)** 110° **(c)** 120° **(d)** 100°

9. **Assertion (A):** The exterior angle of a triangle equals the sum of the two interior opposite angles. 1

Reason (R): The exterior angle of a triangle always equals the adjacent interior angle.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A triangle can have angles 90°, 60° and 40°. 1

Reason (R): The sum of the three angles of a triangle is 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 130^\circ$. Find $\angle A$ and $\angle B$. 2

12. List all isosceles triangles with whole-number sides and perimeter 17 cm (equal sides a, base b). How many are there? 2

13. In $\triangle ABC$ and $\triangle PQR$ it is known that $AB = PQ = 6$ cm, $\angle B = \angle Q = 55^\circ$ and $BC = QR = 8$ cm. Are the triangles necessarily congruent? Give the rule or the reason. 2

14. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 145^\circ$. Find $\angle A$ and $\angle B$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, $\angle A = 70^\circ$. The exterior angles at B and C are $(2x + 75)^\circ$ and $(5x + 70)^\circ$. Find x and the angles B and C. 3

16. The perimeter of an isosceles triangle is 38 cm. Each equal side is 1 cm longer than the base. Find the sides and show that such a triangle exists. 3

17. Two sides of a triangle are 10 cm and 27 cm. The third side is a whole number of cm divisible by 6. List all possible lengths and the largest perimeter. 3

OR

In $\triangle ABC$, $\angle A = 60^\circ$. The exterior angles at B and C are $(6x + 62)^\circ$ and $(2x + 114)^\circ$. Find x and the angles B and C. 3

18. The perimeter of an isosceles triangle is 47 cm. Each equal side is 4 cm longer than the base. Find the sides and show that such a triangle exists. 3

OR

Two sides of a triangle are 9 cm and 22 cm. The third side is a whole number of cm divisible by 3. List all possible lengths and the largest perimeter. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A roof frame is a triangle ABC with $\angle A = 50^\circ$ and $\angle B = 70^\circ$. Side BC is extended to D. 5

(i) Find $\angle C$.

(ii) Find the exterior angle $\angle ACD$.

(iii) Verify that the exterior angle equals $\angle A + \angle B$.

(iv) Name the triangle by its angles and by its sides, giving reasons.

20. The exterior angles of a triangle (one at each vertex) are in the ratio $2 : 3 : 4$. Find the three exterior angles, the three interior angles and classify the triangle. 5

OR

$\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 6)^\circ$, $\angle D = (5x - 24)^\circ$, $AB = (2y + 8)$ cm and $DE = (3y - 1)$ cm. Find x, y, $\angle A$ and AB. If $\angle B = 45^\circ$, find $\angle F$. 5

END OF QUESTION PAPER — Total: 40 Marks